Marine Heat Waves: “The Blob”

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In 2014-15 the eastern Pacific Ocean experienced anomalously warm water known as “The Blob” due to anthropogenic climate change effects on the ocean. Herein we describe the physical and chemical processes that caused these warm temperatures to occur and its consequences for oceanic carbon, oxygen, and ecosystems. Placed in the context of current research findings, we will describe the effects of this marine heat wave, the likelihood that it will occur under an increasingly warming climate, and the potential policy implications that must follow.

Background and Overview: “The Blob”

What are marine heatwaves?

Long periods of very warm Sea Surface Temperatures (SST) that can encompass thousands of kilometers in the ocean. Extreme SSTs are likely caused by atmospheric or oceanic forcings, such as La Nina events, which create higher sea surface pressures.

What is “The Blob”?

The largest-recorded MHW that transpired in the NE Pacific Ocean from 2013-2015. SST anomalies greater than 6°C off the coast of Southern California were recorded. It was attributed to strong positive sea level pressure anomalies, hindering loss of heat from the ocean due to the lack of upwelling water. The blob caused changes in water-circulation patterns, shifted species distributions, and created warmer and dryer weather conditions along the western coast of the United States. The 2015 El Nino is credited with disbanding “The Blob” by bringing lower than normal sea surface pressures to the eastern Pacific Ocean.

How are MHWs being observed and evaluated?

Satellite observations and prediction through Earth system models (ESM).

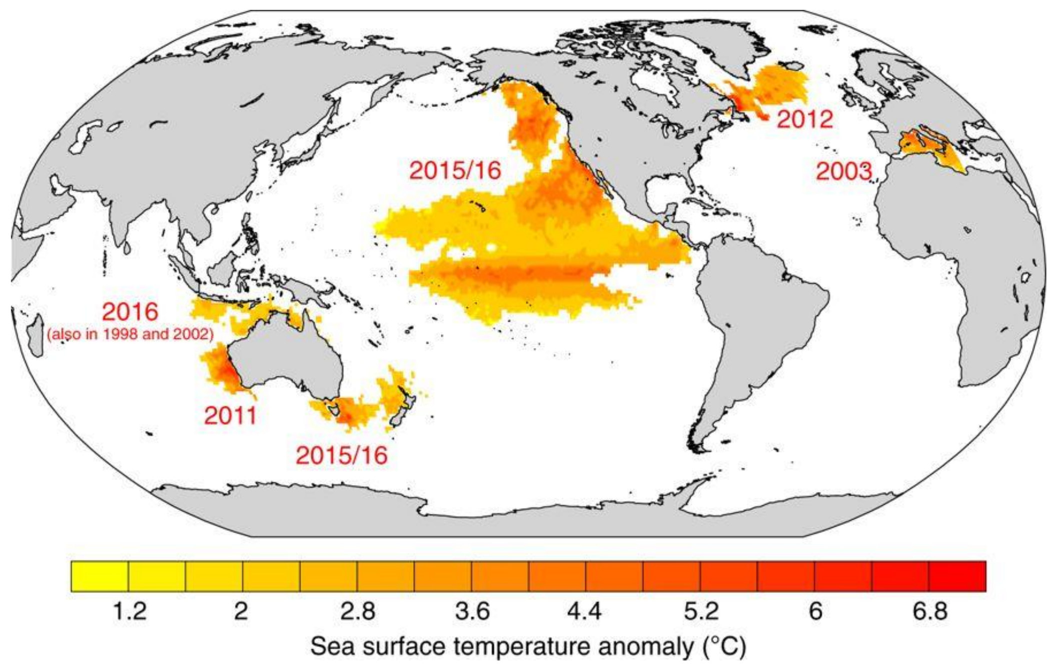

Fig 1: Map of recent and notable MHWs. The figure displays the SST anomalies in each region. Max SST for “The Blob” was ~6°C.

Carbon and Oxygen Consequences

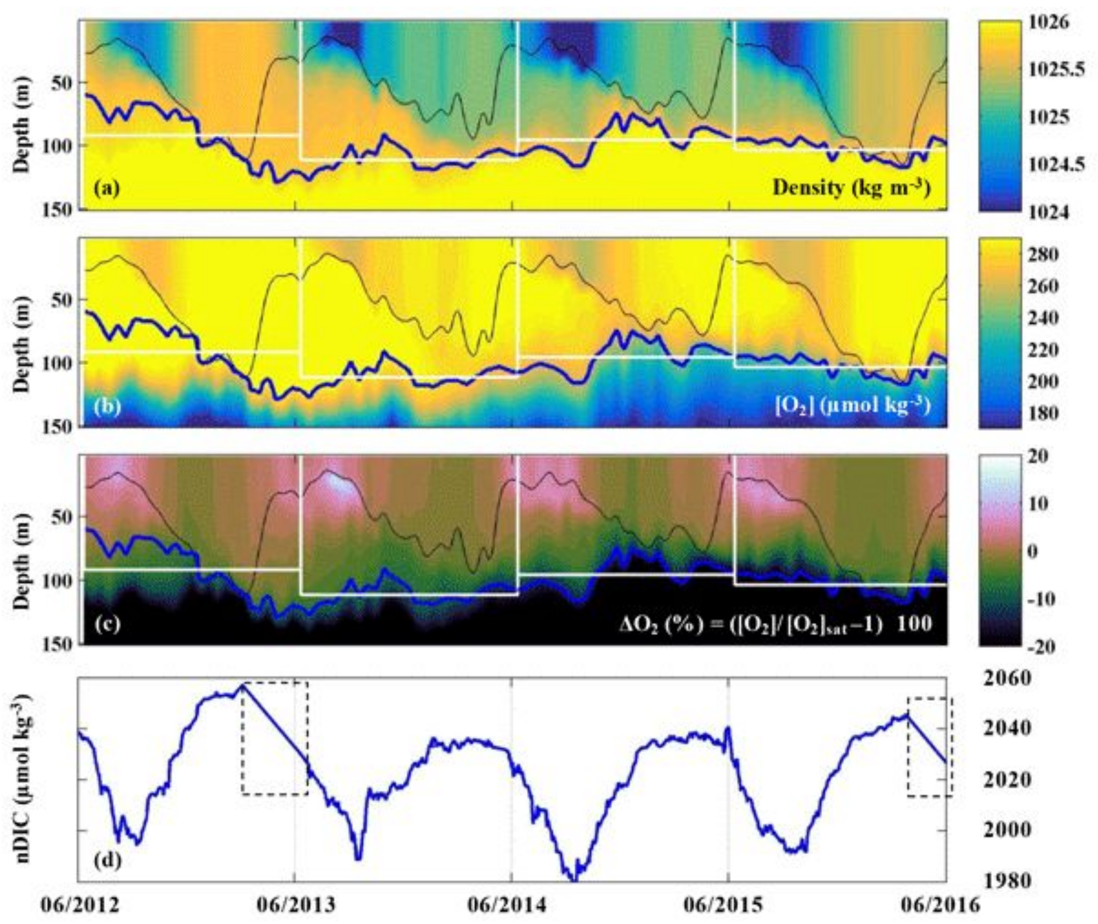

Figure 2 (a-d): Upper ocean density, oxygen concentration, oxygen supersaturation, and mixed layer DIC data measured from 2012-2016 at Ocean Station Papa. (Black line = MLD, Blue line = pycnocline depth).

Carbon : Nitrogen ratio increases when the temperature is at a low and upper extreme of temperature level. This can cause harmful bloom diatoms to grow outside of its optimal thermal range and form blooms (HAB).

Increased SST leads to increase CO2 concentrations which can lead to Domoic Acid production. The Eastern Pacific Ocean goes from a carbon sink to a carbon source to the atmosphere.

During “Pre-Blob” (2012-2013), DIC and Oxygen concentration were high. During “The Blob” (2013-2015), DIC and Oxygen concentration decreased, which caused hypoxic (depleted oxygen) conditions.

Ecosystem Consequences

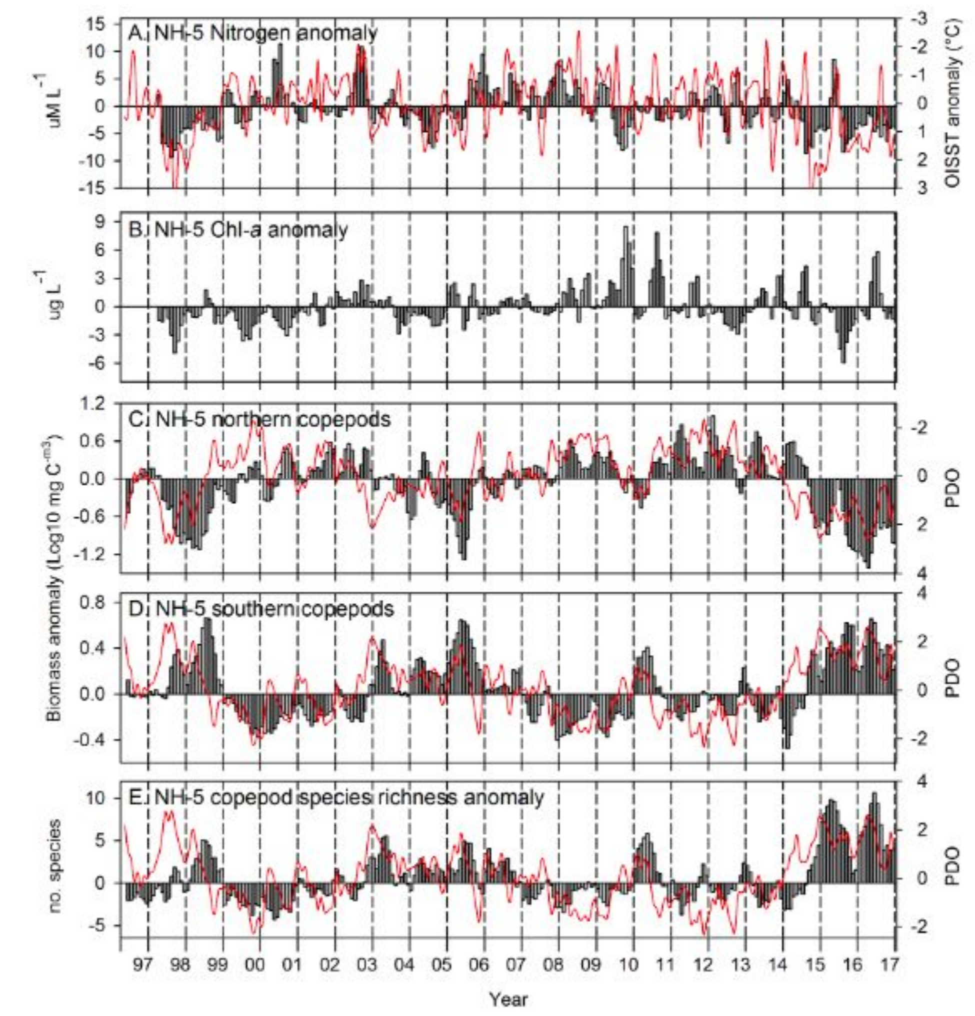

Fig 3: Nitrogen, chlorophyll-a, northern and southern copepod biomass and copepod species richness (vertical bars) anomalies at station Newport Hydrographic Line (NH-5), with Optimum Interpolation Sea Surface Temperature (OISST) at the station and Pacific Decadal Oscillation PDO overlaid as the red lines.

The “blob” lowered primary productivity and altered the composition of the phytoplankton and zooplankton communities. The combined lower productivity and shift to greater abundance of warm-water, lipid-poor zooplankton led to consequences at higher trophic levels. For example, the “blob” caused a die-off of the zooplanktivorous seabirds Cassin’s Auklets and significantly lower settlement of spotted sand bass larvae. An economically (>$50 million in Dungeness crab alone) and biologically destructive harmful algal bloom of *Pseudo-nitzschia australis* occurred during the “blob”’s presence. High levels of domoic acid produced by *P. australis* were likely due to depleted silicate:nitrate ratios. There were also anecdotal reports of southern species spotted in northern waters and other changes in species distribution.

Causes of Anamalously Warm Water in 2014-15

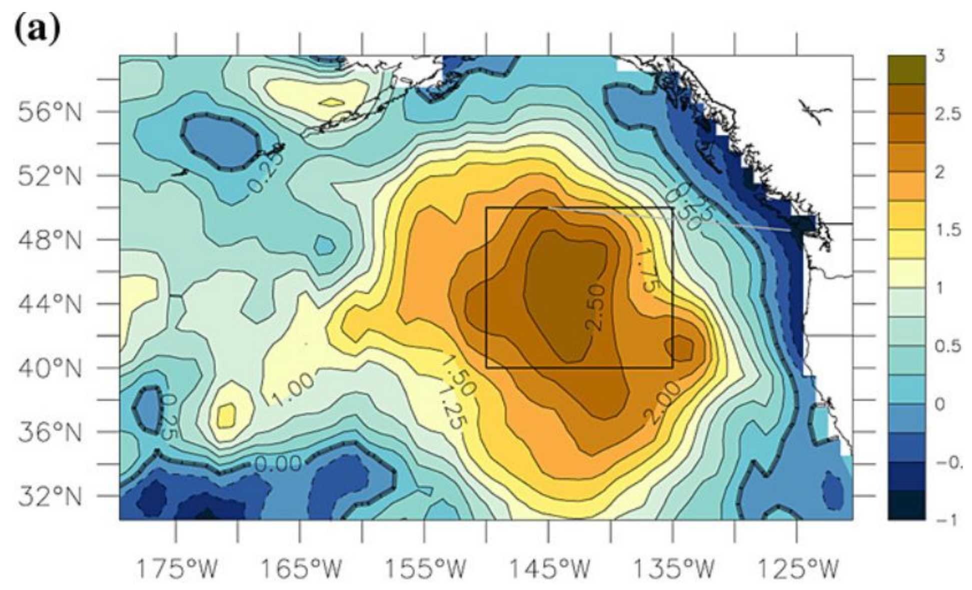

Figure 4(a) Sea surface temperature (°C) in NE Pacific Ocean during February 2014. Deviations from the mean (1981-2010).

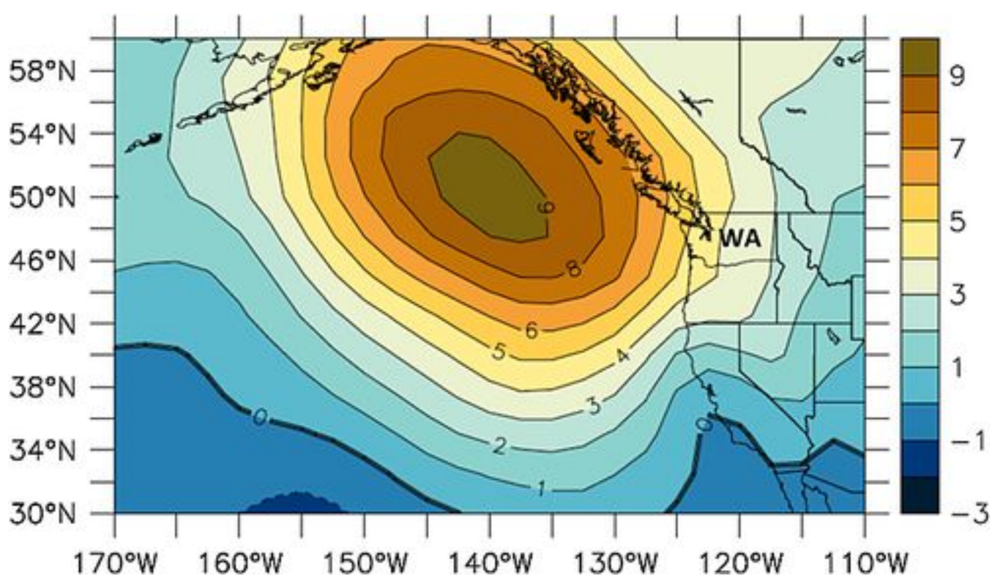

Figure 4(b) Sea level pressure (hPa) in the NE Pacific Ocean during October 2013-January 2014. Deviations from the mean (1981 to 2010).

Two mechanisms account for the anomalously warm water:

1. Unusual decrease in heat loss from the ocean to the atmosphere
2. Weak cold advection in upper ocean

These mechanisms were caused by strong positive sea level pressure. Its effects impacted wind forced currents, wind-generated mixing, and surface heat loss.

Chance of Recurrence

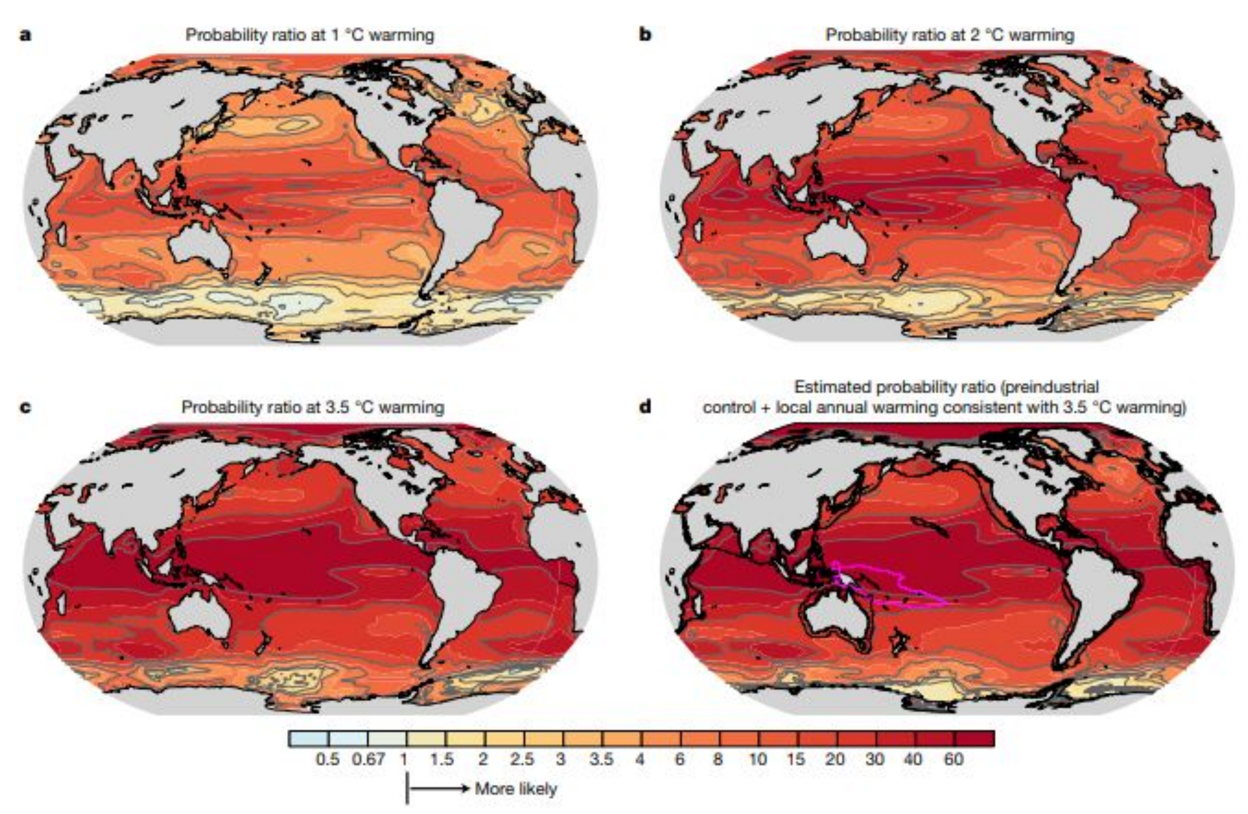

Fig 5 (a-d): Regional changes in the probability of MHW days exceeding the preindustrial 99th percentile for a global warming level of 1 °C (a), 2 °C (b) and 3.5 °C (c). (d) shows that the occurrence of MHWs is mainly driven by a simple shift of the whole temperature distribution, with local annual SST change that is consistent with a 3.5 °C global warming to the preindustrial SST distribution

For the future in all ocean basins, ESMs predict more frequent, extensive, intense and longer-lasting MHWs. For 3.5°C warming, the probability of occurrence of an MHW is 41 times higher than pre-industrial times. The spatial extent of the annual mean is projected to become an averages of 21 times larger, duration increase to 112 days, and maximum intensity to rise to 2.5°C. The probability of occurrence for an MHW under the 1.5°C warming scenario is 40% of that under 3.5°C warming. The probability of MHWs is projected to increase almost everywhere, largest in the tropics and the Arctic Ocean and smallest in the Southern Ocean due to the small variations in SST both seasonally and year to year. In contrast, in the Southern Ocean SST are projected to stay relatively cool, and thus the probability ratio does not increase much under all warming levels.

Policy Implications and Future Directions

From 1925 to 2016, global average MHW frequency and duration increased by 34% and 17%, respectively, resulting in a 54% increase in annual marine heatwave days globally. Research findings strongly link MHWs to climate change. Current national policies for the reduction of global carbon emissions are predicted to result in global warming of about 3.5 degrees Celsius by the end of the twenty-first century. MHWs can cause widespread loss of kelps and corals, shifts species distributions, alter communities and ecosystem structures, and have economic impacts on aquaculture and seafood industries through declines in important fishery species. The warm blob in the northeast Pacific caused the closing of both commercial and recreational fisheries resulting in millions of dollars in losses among fishing industries. Policy solutions can include:

1. **Enhance direct incentives for mitigation.** Introduction of new policies and instruments that shape the incentives for economic actors in favor of actions that limit GHG emissions.
2. **Mechanisms for mainstreaming.** Explicit framework for integrating and incentivizing energy, transport, and urban planning to address climate change. Mainstreaming a unified policy objective can help achieve more than disconnected sectoral policies can accomplish in isolation.
3. **Collaboration between bureaucracies, NGOs, and the private sector.** Providing opportunities to the private sector and NGOs will help civil society organize and induce normative shifts in favor of effective mitigation action.

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